This is a draft of the coursework offered in advance of the coursework period; there may be some adjustments between now and then.

Coursework

This coursework has two parts. Parts I (25 marks) contains computational modelling and simulation questions based on the course labs. Part II (25 marks) explores statistical analysis of spiking data. Each part contains 3 questions worth 5 marks each, and 1 open-ended question worth 10 marks, for 50 marks total toward your final grade.

Part I: Computational Modelling

This section contains 4 questions extending the labs (with the exception of perceptron and delta rule, which are touched on in part II).

Question 1 integrate-and-fire [5 marks]

Consider the Quadratic Integrate and Fire (QIF) neuron with dynamics $\tau_m \dot{v} = -v(1-v) + R_m I$. This model "spikes" by blowing up to $+\infty$ (and "resetting" through $-\infty$) in finite time. In practice, numerical simulations set some finite positive voltage for the threshold, and resets to rest (Choose these thresholds at your discretion). [1]

(i) Simulate the QIF model[2]. Explore some inputs where the model does, and does not spike, and explore how it integrates and responds to inputs near threshold. Use plots from this exploration to support your analysis in part (ii).

(ii) Compare the QIF firing behaviour to that of the Leaky Integrate-and-Fire (LIF) neuron. What qualitative features does the QIF model capture that the LIF cannot?

Question 2 conductance models [5 marks]

The phenomenon of *post-inhibitory rebound spiking* occurs in some neurons when a hyperpolarising current pushes a cell's voltage below its resting voltage. This opens or deinactivates voltage-gated channels, making the cell more excitable. As a result, a neuron that would normally be silent at rest can emit a spike when the inhibition is withdrawn.

(i) Demonstrate this phenomenon using one of the Hodgkin-Huxley models from lab 3: Find a current input that hyperpolarises the cell, and elicits a rebound spike when withdrawn. Communicate the results of your simulation with the aid of figures.

(ii) Discuss the role of voltage-gated sodium channels in post-inhibitory rebound spiking in this model, and discuss this in context of the idea of a *threshold* (do conductance-based models really have a threshold)?

[1] You can also change coordinates map this "voltage" to a circular variable θ, with θ passing through π corresponding to the 'spike' up to and through $+\infty$ and back around from $-\infty$, allowing you to simulate the equations without choosing arbitrary finite thresholds.

[2] One way to do this is to use forward Euler, handling the reset manually.

Question 3 Synapses [5 marks]

Consider a glutamatergic synapse where a presynaptic spike at time $t = 0$ instantly releases concentration q_0 of glutamate into the synaptic cleft, which binds (reversibly) to post-synaptic AMPA receptors (AMPARs) and is cleared away by reuptake and diffusion[3]. Let $q(t) \geq 0$ be the concentration of free glutamate in the synaptic cleft, initially $q(t{=}0) = q_0$, and $s(t) \in \{0, 1\}$ be the fraction of post-synaptic AMPA receptor binding sites bound to glutamate.

Model 1: A normalized mass action chemical kinetics model obeying the nonlinear ODE

$$\dot{s} = -k_d\, s + k_b\, q(1 - s) \qquad \dot{q} = -k_c\, q - \dot{s}\mathrm{A}_{\mathrm{total}}\,. \tag{1}$$

Model 2: Model 1 generally lacks nice closed-form solutions. For some parameters we can use two linear first-order ODEs as a good approximation:

$$\dot{s} = -k_d s + k_r q, \qquad \dot{q} = -k_c\, q, \tag{2}$$

which for $s(0) = 0$, $q(0) = q_0$, $k_d \neq k_c$ has a difference-of-exponentials solution $s(t) = k_r q_0 \frac{e^{-k_c t} - e^{-k_d t}}{k_d - k_c}$.

Parameter	Value	Units	Description
A_{total}	0.32	mM	Stoichiometry conversion factor
k_d	0.19	ms^{-1}	Dissociation rate constant
k_b	1.1	$\mathrm{ms}^{-1}\mathrm{mM}^{-1}$	Binding rate constant
k_r	0.67	$\mathrm{ms}^{-1}\mathrm{mM}^{-1}$	Effective gating rise rate constant
k_c	5	$\mathrm{ms}^{-1}\mathrm{mM}^{-1}$	Rate constant for glutamate clearance

Table 1: *Synapse model 1 and 2 parameters. These are effective parameters (mM is millimolar, but do not worry much about units: Use these as-is to simulate in time units of milliseconds).*

(i) Simulate ODEs (1) and (2) using the parameters in Table 1, and $q_0 = 1$ mM (they should be similar). These simulations should be included, discussed, and referenced, somewhere in a plot (or part of a plot) for your overall answer to this question.

(ii) Compare models 1 and 2 for a range of release amounts (q_0), say from q_0=0.1 to q_0=10. Where do the models agree, differ, and when does it matter? Interpret these models in the context of physiology and alternative synaptic models.

[3] You may find *Ermentrout and Terman* chapter 7 helpful, the textbook is on the course Github.

Question 4 Hopfield [Open-ended — 10 marks]**

In the Hopfield attractor network lab, you may have noticed that the network had poor capacity: It struggled to store all 16 hexidecimal digits distinctly and without spurious attractors.

(i) (open ended) Extend your Hopfield solution to mimic dentate gyrus' *pattern separation*: You should expand the given binary patterns into larger, sparse binary vectors, and show that these expanded representations can be stored more effectively. Your pattern separation routine should be a deterministic function of the original patterns, and you should comment on how your approach does (or does not) relate to the dentate gyrus' approach of using a (~random) weight matrix to expand patterns into a higher-dimensional space with k-winner-take-all sparsity.

***This is an open-ended question designed to allow students to demonstrate study and originality beyond what has been taught, for those aiming for top marks.*

Part II: Statistical Data Analysis

Part II gives a data-led perspective on spike trains. Some questions are adapted from part 1.1 of the Dayan and Abbott textbook. The spike-train statistics coursenotes accompany this section.

Question 5 Poisson, Fano, and CV [5 marks]

This question is about Poisson processes.

(i) First, write code to generate spikes using a Poisson process and sample a homogeneous Poisson spike train with firing rate of 35 Hz. Then, extend your code to sample from an inhomogeneous Poisson neuron that has a 5 ms absolute refractory period, keeping the overall firing rate at 35 Hz (35 spikes per second). Briefly describe what this code does using mathematics and figures.

(ii) Plot how the ISI coefficient of variation and spike-count Fano factor in 100 ms bins change over refractory periods ranging from 0 to 28 milliseconds. Include a descriptive figure title and a caption with enough detail to understand the context of what is being plotted and the effect that the figure is demonstrating.

Question 6 Data Inspection [5 marks]

In the folder for this extended coursework, you will find the data file `rho.dat`. This contains data collected by Rob de Ruyter van Steveninck from a fly H1 neuron responding to an approximate white-noise visual motion stimulus. Data were collected for 20 minutes at a sampling rate of 500 Hz. In the file, rho is a vector that gives the sequence of spiking events or non-events in 2 ms time bins.

(i) Calculate the Fano factor and coefficient of variation for this spike train as for the simulated spike trains above (also exploring 10 ms, 50 ms, 100 ms bin size). Does the spike train exhibit more variability or less variability than you'd expect for a homogeneous Poisson process? How is this reflected in the count Fano factor and ISI coefficient of variation? Conjecture on why the data may deviate from a homogeneous Poisson process in this experiment. (25-100 words)

Question 7 Spike-Triggered Average (STA) [5 marks]

In the folder for this extended coursework you will find the data file `stim.dat`. This gives the motion stimulus that evoked the spike train in `rho.dat`.

(i) Calculate and plot the spike triggered average over a 100 ms window.

(ii) Calculate the stimulus triggered by consecutive pairs of spikes: For intervals of 2 ms[4], 4 ms, 10 ms, and 20 ms calculate the average stimulus before a pair of spikes separated by that interval. Visualise your results and discuss the role of the refractory period in what you see.

[4] i.e. two adjacent 2 ms bins each with a spike

Question 8 Perceptron → Poisson [Open-ended — 10 marks]**

Learning rules similar to the delta rule appear when training statistical models of neural encoding. We will explore a linear–nonlinear Poisson encoding model using the fly H1 data (inputs u from `stim.dat`, predicting spikes y from `rho.dat`):

$$\begin{aligned} \Delta \boldsymbol{w} &= \eta \cdot \boldsymbol{x}\varepsilon \\ \varepsilon &= y^* - \lambda \\ \lambda &= \exp(\boldsymbol{w}^\top \boldsymbol{x}). \end{aligned} \tag{3}$$

For each time bin i, define a non-negative integer spike count y_i and input feature vector $\boldsymbol{x}_i^\top = \{1, u_{i-1}, u_{i-2}, \ldots, u_{i-K}\}$ which includes a bias (threshold) term and stimulus history for K time steps. With this setup, the rule (3) will learn weights $\boldsymbol{w}^\top = \{-\vartheta, w_{u_{-1}}, \ldots, w_{u_{-K}}\}$ that contain a (negative) threshold and stimulus history filter.

(i) Write code to learn $\boldsymbol{w}$ for a history length $L = 100$ bins[5]. You may modify your perceptron lab, or use another machine learning (ML) solution. (You will need to prepare the feature vectors $\boldsymbol{x}$ yourself from the H1 data). Without regularisation, your history filter may alternate between positive/negative numbers[6]—this is expected. Show how well the trained model predicts neural activity[7].

(ii) **(Removed, see next page)** One way to regularise is to add zero-mean Gaussian noise to the stimulus during training[8]. Re-learn $\boldsymbol{w}$ using regularisation on the history filter[9]. Comment on the relationship between the history filter and the Spike-Triggered Average (STA), and on physiological interpretation of noise for learning in the brain[10].

(iii) Extend your model of the stimulus → activity encoding to capture refractory effects by making it an *autoregressive* point process, by creating history features that depend on the past of the spike-count time series $\boldsymbol{y}$. One way to achieve this with regularization is to filter $\boldsymbol{y}$ through a handful of exponential filters akin to synapses with various time constants[11].

This is an open-ended question designed to allow students to demonstrate study and originality beyond what has been taught, for those aiming for top marks. Some students may not complete part (iii).

[5] If using the delta-like rule, try $\eta = 10^{-6}$. You may want to wrap exp() to avoid numerical overflow.
[6] The stimulus lacks variation at high frequencies, so learning over-fits to these details.
[7] For example by comparing the predicted λ and true y in 10 ms bins (or another graphic of your own design)
[8] In our test, σ^2=1/sample worked well, i.e. `x_i=concatenate([[1],u[i-L:i]+randn(L)])` in pseudo-numpy.
[9] You do not need to regularise the threshold parameter.
[10] If you chose another approach, comment on its connection to the delta-like rule & regularisation via noise.
[11] Models with only two exponential filters can work well, but unless you are using a ML solution that can tune time constants, you may want to fit 5-10 time constants at the same time, to see which ones the model uses. Raised-cosine basis functions that average over progressively more time further into the past are another popular choice

Question 8 bug fixes

2025.11.25: Dear all: I made a mistake attempting to "simplify" the regularisation approach to the GLM fitting. That said, the over-fit (un-regularised) model should still show good prediction on this dataset (which is remarkably stationary). **For fairness, this is removed from the coursework entirely (you will not lose marks for anything related to Q1ii).**

Students do not need to know the remainder of this page for the SEMT30003/4, this derivation is provided only for transparency, completeness. Let's derive another way to regularise the stimulus history filter (the approach of projecting onto a smaller set of smooth basis functions is already mentioned in the context of the spike history filter in Q8iii). For this, it will help to keep the threshold parameter ϑ and history weights separate. To avoid confusion with the notation in Q8, define the stimulus history $\boldsymbol{z}_i^\top = \{u_{i-1}, u_{i-2}, \ldots, u_{i-K}\}$ and call the history weights $\boldsymbol{m}^\top = \{w_{u_{-1}}, \ldots, w_{u_{-K}}\}$. We will take a Bayesian approach combine our Poisson likelihood of y_i given weights $\boldsymbol{m}$ with a multivariate Gaussian prior encoding the assumption that $\boldsymbol{m}$ should be smooth.

$$\begin{aligned} y_i &\sim \text{Poisson}[\lambda_i = \exp(\boldsymbol{m}^\top \boldsymbol{z} - \vartheta)] \\ \boldsymbol{m} &\sim \mathcal{N}(\mu_m{=}0, \Sigma_m). \end{aligned} \tag{4}$$

Dropping terms constant in $\boldsymbol{m}$ (which don't affect optimisation), the log-likelihood, prior are:

$$\begin{aligned} \ln \Pr[y_i|\boldsymbol{z}_i] &= y_i(\boldsymbol{m}^\top \boldsymbol{z}_i - \vartheta) - \exp(\boldsymbol{m}^\top \boldsymbol{z}_i - \vartheta) + \text{constant} \\ \ln \Pr[\boldsymbol{m}] &= -\tfrac{1}{2}\boldsymbol{m}^\top \Sigma_m^{-1} \boldsymbol{m} + \text{constant}. \end{aligned} \tag{5}$$

We'll maximise the log-posterior, which is the sum of the average log-likelihood and log-prior:

$$\begin{aligned} \mathcal{J}(\boldsymbol{m}, \vartheta) &= \langle \ln \Pr[y_i|\boldsymbol{z}_i] \rangle_i + \ln \Pr[\boldsymbol{m}] \\ &= \langle y_i(\boldsymbol{m}^\top \boldsymbol{z}_i - \vartheta) - \exp(\boldsymbol{m}^\top \boldsymbol{z}_i - \vartheta) \rangle_i - \tfrac{1}{2}\boldsymbol{m}^\top \Sigma_m^{-1} \boldsymbol{m} + \text{constant}. \end{aligned} \tag{6}$$

We maximise (6) by gradient *ascent*. Considering just the gradient of (6) in $\boldsymbol{m}$:

$$\Delta \boldsymbol{m} \propto \nabla_{\boldsymbol{m}} \mathcal{J}(\boldsymbol{m}, \vartheta) = (\ldots) - \Sigma_m^{-1} \boldsymbol{m}, \tag{7}$$

Where $(\ldots)$ are terms from the expected Poisson log-likelihood in that students should derive.

What do we choose for Σ_m^{-1}? Our goal is to remove high frequencies from $\boldsymbol{w}$ that aren't available to learn from data. Many reasonable choices can accomplish this. Here is a fun one: We could take inspiration a heat/diffusion[12] equation $\dot{\boldsymbol{q}} = \gamma \nabla^2 \boldsymbol{q}$ on the weights (with history time lag acting as a sort of spatial variable, and γ setting diffusion rate), implying $\Sigma_m^{-1} = -\gamma \nabla^2$. If we ignore the $(\ldots)$ terms, a discrete step has a solution in terms of the matrix exponential $\boldsymbol{m}_{k+1} = \exp(-\gamma \nabla^2)\boldsymbol{m}_k$, which corresponds to applying a small Gaussian blur (convolution of $\boldsymbol{m}$ with a Gaussian kernel) to the weights on each training step. This regularises high frequencies and attenuates variance in the weight estimates. Thus, the intuitive *"simply blur out the weights a bit while your train"* approach to keeping $\boldsymbol{m}$ smooth has a principled interpretation as a Bayesian prior.

[12] look up the discrete Laplace operator for how to write ∇^2 as matrix/vector operations.

Report instructions

Submit your coursework as a professionally formatted document in the `.pdf` file format. Do not use a font size less than 11 pt and have normal looking margins. Single-spaced text is preferred. Organise the report with the same structure as this document, with question and sub-questions numbers as headers and sub-headers, respectively. Any coding language can be used, and you should include small code extracts in the text where appropriate. You will also be required to upload your code. Submission is through blackboard upload.

There are 50 marks in total, this will be used to calculate a percentage. The two questions marked with an asterisk are harder and more open ended; do not spend too much time on these, the marking will give some credit for any attempt at the modelling they involve, but to get most or all of the 15 marks will require some experimentation.

Scientific communication relies strongly on figures. Communicate your work using professionally prepared figures and tables throughout. Marks will be deducted for graphs where the text is tiny, or relevant units, titles, or axis labels are missed. Do not include graphs that have been produced by screenshots. Neuroscience as a subject favours long figure captions that provide detailed information about what the graphs show, without giving interpretation (that is left to the main text). Follow this convention. It is easier to design consistent clear figures if you set the figure size in code to consistently match the text width. You do not need to place every plot for every separate sub-question (i,ii,iii) as its own figure. Please consider designing a single, tasteful, compound figure (i.e. with sub-figures) to reference throughout your answers to each question.

See if you can cover all points in as few as six pages. This is not target length a much shorter submission could still earn all or most of the marks. Exceptionally, seven-page submissions may be accepted, but please check with course instructors to ensure you are not doing more work than expected. The reason the length is not shorter is to make it easier to write without having to worry about being very precise—not because a long response is required.

Your submission should contain professionally typeset mathematical expressions. You will find it easier to achieve a professionally formatted result if you use the document preparation system LaTeX. The University of Bristol maintains licenses for Overleaf.

Use of Artificial Intelligence

This assessment counts towards your final grade and the default University of Bristol AI policy ("you must not use AI") is in effect, with the following exceptions: You may use the AI tools to check grammar, debug LaTex, and prepare LaTex source code for tables and figures. This coursework assesses judgement, taste, and insight in scientific communication, not how quickly you can debug LaTex errors.

Marking Criteria

See the separate `marking_criteria_extended_coursework_SEMT30003.pdf` in this coursework folder for details of how the university marking criteria will apply to your submission.

Moderation

Moderation of coursework will be performed by a second marker as follows:

Let N be the number of total coursework submissions. A random sample of $\max\{8, \text{ceil}(N/10)\}$ submissions distributed across the grade boundaries (fail/pass/merit/distinction) will be drawn and reviewed.

Possible outcomes from moderation are:

- Moderator confirms marks (no changes)
- Moderator adjusts all marks to improve alignment with the marking criteria and/or align the mark distribution to program norms[13].
- A subset of marks may be adjusted to fix inconsistencies.
- Some or all assessments may be re-marked if inconsistencies cannot be fixed easily.

[13] This is not quite the same as "we will curve the overall cohort marks to match UK grade boundary norms", but should provide you with a similar level of reassurance. A year with unusually poor or unusually excellent submissions will not be inflated or punished on account of cohort performance. Rather, the relationship between quality and mark will be moderated based on overall performance from previous years, and across similar courses. This is a sort of spatiotemporal-averaged curving mechanism, with the added benefit that ongoing audits and reviews of program difficulty allow this process to better reflect a normative consensus for UK degree programs, rather than allowing larger per-cohort fluctuations to bias the individual quality→mark mapping like a within-cohort curve would.